Víctor Rincón Yepes

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Summary

Computer Vision and Machine Learning Engineer with **4+ years** of professional engineering experience, focused since 2024 on production deep learning for very high resolution aerial imagery (5-10 cm GSD) at **global scale**. At Vexcel, serving as **senior technical reference** within a 6-person computer vision team, driving architecture decisions across instance and semantic segmentation, object detection, property damage assessment, and remote sensing analysis. Models deployed across North America, South America, Europe, Oceania, Japan, and South Africa, covering millions of properties in a big data environment.

Experience

Senior Computer Vision and Deep Learning Engineer *Dec 2024 - Present*
Vexcel Group (Vexcel Data Program), Madrid, Spain

- Senior technical reference within a 6-person computer vision engineering team, contributing to architecture decisions and technical direction across parallel research and production projects.
- Designed a bifurcated multitask architecture (Depth Anything V2 backbone with normalized distance transform regression) for instance-level building footprint segmentation, achieving **0.94 mIoU** and **0.93+ Object IoU**, a **+5 point improvement** over the previous production model, currently **deployed in production**.
- Led the development of the post-disaster property damage assessment pipeline for insurance claims and emergency response from aerial imagery, improving F1 score **from 0.50 to 0.80**; currently deployed as a **revenue-generating product**.
- Primary technical point of contact for **two external annotation vendors**, defining labeling guidelines and quality standards for aerial imagery datasets.

Computer Vision and Deep Learning Engineer *Jan 2024 - Dec 2024*
Vexcel Group (Vexcel Data Program), Madrid, Spain

- Designed and trained task-specific deep learning models (semantic and instance segmentation networks, object detectors, and regression models) on large-scale aerial orthoimagery, leveraging CNN backbones (ResNet, EfficientNet, HRNet) and vision transformers (ViT, Swin), targeting diverse property features such as solar panels, chimneys, AC units and damage indicators.
- Benchmarked and adapted state-of-the-art segmentation architectures (SegFormer, DeepLab, HRNet, SAM3, DINOv2, among others) to the geospatial aerial imagery domain, establishing production baselines.
- Implemented **active learning** pipelines for aerial image quality control, selecting high-uncertainty samples via model behaviour analysis and routing them for annotation in CVAT, reducing manual labeling overhead.

Software Engineer *Aug 2022 - Jan 2024*
Vexcel Group (Vexcel Data Program), Madrid, Spain

- Technical lead for core backend geospatial services, owning full-cycle development from requirements analysis to production deployment, following Scrum/Agile methodologies.
- Built scalable backend web services and REST APIs (Java Spring Boot, JPA, Hibernate) integrated with geospatial data pipelines, with deployments on Kubernetes and AWS (Lambda, S3, SQS) for processing and storage.
- Designed and maintained spatial databases (PostgreSQL/PostGIS) for large-scale property data, including complex geometry manipulation and GIS processing.

Software Development Intern *Jan 2022 - May 2022*
Vexcel Group (Vexcel Data Program), Madrid, Spain

- Assisted in the development of web applications and internal tools using Java and Angular.

Education

Master's Degree (MSc) in Artificial Intelligence *Sep 2023 - Jun 2025*
Universidad Internacional de Valencia (VIU), Spain

- Thesis: *Medical Image Diagnosis with Vision Transformers*, a comparative study of CNN vs. ViT architectures on dermatoscopic classification (ISIC 2020), including a pruning and quantization pipeline for edge deployment. Swin Transformer outperformed modern CNNs while maintaining computational efficiency.

Bachelor's Degree (BSc) in Telecommunications Engineering *Sep 2017 - Jul 2022*
Universidad Rey Juan Carlos (URJC), Madrid, Spain
Specialization in Telematics Engineering

Publications & Technical Writing

- *Phylogenetic signal in marine mammal and bird vocalizations captured by audio foundation models: the limited benefit of domain-specific pretraining*, arXiv preprint, 2026. Code and data. Foundation-model audio embeddings recover phylogenetic structure across marine mammals and birds; domain-specific pretraining does not help.
- *Bifurcated Boundary-Aware Depth Transformer for Instance-Level Footprint Segmentation*, in preparation, arXiv 2026.
- *Faster than Maps: How SAR Satellites Expose the Blind Spots of Commercial Cartography*, GeoAwesome, 2026. Proposes a hybrid SAR and GeoAI monitoring pipeline to overcome cartographic lag in rapidly urbanizing regions.

Technical Skills

Programming Languages: Python, Java, SQL, Bash, TypeScript/Angular (basic).

Machine Learning & Computer Vision: PyTorch, HuggingFace Transformers, OpenCV, scikit-learn, NumPy, Pandas, semantic segmentation, instance segmentation, object detection, image classification, vision transformers (ViT, Swin, SegFormer, DINOv2), CNN architectures (ResNet, EfficientNet, HRNet, DeepLab), foundation models, transfer learning, active learning, data augmentation, model pruning and quantization, model evaluation and benchmarking.

MLOps & Tooling: MLflow (experiment tracking), Airflow, CVAT, Docker, Kubernetes, Jenkins (CI/CD), Git, Linux, Jira, annotation and data pipelines.

Remote Sensing & GIS: QGIS, GDAL, Rasterio, Shapely, PostGIS, GeoAI, aerial orthoimagery and orthomosaics, SAR, raster and vector processing, coordinate reference systems.

Backend & Cloud: Java Spring Boot, JPA/Hibernate, REST APIs, PostgreSQL, Flyway, AWS (S3, EC2, Lambda, SQS).

Spoken Languages

Spanish: Native

English: Full Professional Proficiency (C1 level)